RYAN RUNDLE

rdrundle@outlook.com | (714) 403–7837 |ryandrundle.github.io/web/

EDUCATION

University of California, Berkeley – B.S. in Mechanical Engineering, GPA**:** 3.5/4.0 Dec 2025

SKILLS

Design & Analysis: SolidWorks, PDM, Onshape, Ansys, Fusion360, GD&T
Programming: MATLAB, Python, C++
Manufacturing & Testing: Machining (lathe, mill), 3D Printing (FDM, SLA), Laser Cutter, FT-IR, XRD, TGA

EXPERIENCE

Research Affiliate, Liwei Lin Lab & Lawrence Berkeley National Lab – Berkeley, CA Link ↗ Oct 2024 – September 2025

- Designed a square hydrogel curing mold to replace single-sample petri-dish curing, eliminating sample damage from manually cutting gels to square profile and enabling simultaneous curing of 9 samples per mold
- Developed characterization protocols for 3 different types of hydrogels by researching optimal preparation methods and instrument parameters for spectroscopic, structural and thermal analysis with FT-IR, XRD, and TGA
- Validated and analyzed experimental data by cross-referencing results with databases, textbooks and similar publications, interpreting 3 key diagnostic features and writing explanations of underlying physiochemical phenomena

Mechanical Engineering Intern, Attollo Engineering – Camarillo, CA May 2024 – Aug 2024

- Created a compact and portable laser target system in SolidWorks featuring a telescoping fold design and deployable wheels, replacing previous temporary target for use of testing laser products in the field and at conferences
- Built target using T-slotted extrusions, iterating through 5 different designs to achieve results of vertically supporting target during windy conditions, housing all reflective hardware internally, and capability of one person setup
- Designed and assembled 7 optical component fixtures on a breadboard with collimation capability, grounding, and heat sinking, allowing for repeated signal transmission and return to test component performance
- Developed a user-safe optical enclosure with 4 microswitches to turn off lasers when doors are open, housing 3 component mounts designed using DFM principles for CNC manufacturing

PROJECTS

UC Berkeley Formula SAE – Brakes and Driver Interface Team Sep 2025 – Present

- Conducted thermal and structural FEA in Fusion360 to optimize front and rear brake rotors, resulting in 28% increase in heat dissipation at max. operating temperature while maintaining required FoS
- Optimizing a new brake pedal using Ansys structural optimization and analysis, cutting unsprung weight and improving knowledge transfer of pedal mechanics

MotoStudent of California – Student Engineering Team Feb 2025 – Apr 2025

- Eliminated the need for 3 separate jigs and saved $200 in tooling by creating a modular welding jig in Onshape with reconfigurable components that hold and position pipes for motorcycle chassis, subframe, and swingarm fabrication
- Designed jig using a t-slotted frame and simple hardware (nuts, bolts, screws, split pipe clamps, sheet metal) that was easy to buy off-the-shelf and required minimal assembly

Rooftop Fire Suppression System – ME102B: Mechatronics Design Link ↗ Jan 2025 – May 2025

- Designed the mounting plate, arms, solenoid holder, and pitch-axis transmission system of a water turret in Onshape that mounts to a residential roof to fight nearby fires
- Iterated through 3 designs to ensure heavier parts were located in the centroid of the turret without interfering with wiring or tubing, resulting in negligible encoder drift and contributing to accurate targeting during demonstrations

Portable Solar Phone Charger – ME100: Electronics for IoT Link ↗ Jan 2024 – May 2024

- Built a self-powered solar tracker out of laser cut plywood using 2 servo-driven panels controlled by an ESP32 microcontroller that successfully delivers 5V output as an off-grid phone charger
- Developed a timer-based program in Python that coordinates solar tracking, humidity sensing, and voltage monitoring with MQTT data transmission, providing user alerts for both weather conditions and 7W solar power availability